# Design System & UX

polycarpic — personal budgeting & wealth management (virtual Single Family Office)

**Owner:** `ux-designer` agent **Status:** Draft — Plan phase, matured alongside ARCH **Version:** 0.14

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# 1. Overview & Design Principles

*What this product should feel like to use, and the principles that resolve design disputes.*

- **Mobile-first when in doubt.** The primary v1 target is a self-hosted web dashboard used on desktop for deep review (statements, lot tables) and on mobile for quick checks (dashboard glance, confirming a draft transaction) — design the dense screens up from a mobile layout, not down from desktop.
- **Accessible by default (WCAG AA).** Non-negotiable per PRD §6. Financial gain/loss and status signals never rely on color alone — see Financial semantics.
- **No decorative loading states.** Skeletons match the real layout; spinners are a last resort (see Component inventory).
- **Every number earns trust.** This is a general ledger, not a budgeting toy — a balance, a delta, or a pending-liability figure must always be traceable to its source (the journal entry, the lot, the reconciliation). The UI never shows a number it can't explain if clicked.
- **Immutability is visible, not just enforced.** A posted transaction reads as posted (badge, disabled edit affordance); a correction reads as a reversal + new entry, never as a silent edit (PRD §5).
- **Diverge from polycarpic's own tooling, deliberately.** The cairn tracker (our internal tool) and the product are not the same thing and shouldn't look like it — this system's identity is derived from **measured** screenshots and live style reads off Monarch Money and Origin (`useorigin.com`), not memory — see Inspiration: Monarch & Origin.
- **One brand hue, used honestly.** The neon-green brand color (`#2CFF05`, "the project name relates to fruiting plants") measures 1.36:1 against white, far below AA — kept as the literal `--brand` value everywhere it clears AA (dark surfaces), with a darkened same-hue green for light-mode `--primary`, stated plainly rather than silently swapped for a different hue — see Foundations.

## 2. Design System

*The single source of truth for the visual language. The written spec lives alongside this doc; the machine-readable system lives in the two CSS files that `frontend-lead` consumes directly.*

- design-system-spec.md § Inspiration — what was taken from Monarch Money and Origin, and why
- design-system-spec.md § shadcn-svelte reuse map — every component/region mapped to its shadcn-svelte component, block, or a stated "custom" reason
- design-system-spec.md — the written spec (token rationale, financial semantics, component inventory, accessibility)
- tokens.css — design tokens as CSS custom properties (hand-authored palette, not a vendored shadcn preset)
- tokens.html — every token decoded: name, value in both modes, swatch, one-line "used for"
- screen.css — component & screen styles built on the tokens

*Figma stays the high-fidelity design tool for anything shipping pixel-exact (via the `ux-designer`'s Figma skills); `docs/DESIGN/` is the in-repo, code-level home that ships with the project and is the contract for implementation.*

# 3. Design Tokens

*Color, typography, spacing, radius, and shadow primitives — SvelteKit + shadcn-svelte (PRD §8). Edit values in tokens.css; the swatches below preview the live variables. Full rationale in design-system-spec.md — or see tokens.html for every token decoded with both-mode values and a "used for" line.*

## 3.1 Surfaces & Text



`--background` `--foreground` `--card` `--popover` `--muted` `--muted-foreground`

## 3.2 Actions & States



`--primary` `--brand (dark surfaces only — see Foundations)` `--brand-2 (same rule — see Secondary accent)` `--secondary` `--accent` `--destructive`

## 3.3 Financial semantics (new — polycarpic-specific)



`--positive` `--warning` `--destructive (reused for negative/danger)`

*See design-system-spec.md § Financial semantics — gain/loss, safe-harbor status tiers, and the "never color alone" rule.*

## 3.4 Borders & Focus



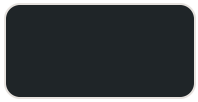
`--border` `--input` `--ring`

## 3.5 Charts (sequential ramp of --brand-2 — magnitude, not category)

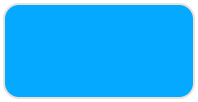


`--chart-1` `--chart-2` `--chart-3` `--chart-4` `--chart-5`

### 3.6 Sidebar (inverted, blue-tinted)


`--sidebar`


`--sidebar-accent`


`--sidebar-ring`

*Dark-on-light in light mode, light-on-dark in dark mode — see design-system-spec.md § Sidebar for the collapse-to-rail behavior and three proposed expand/collapse cues.*

***Type:*** `--font-sans` *Inter — one grotesque sans for body and headings, hierarchy by weight, confirmed against both reference products' actual app UI (their serif accent is marketing-only, see Inspiration);* `--font-mono` *Roboto Mono — Origin's own measured choice, for uppercase eyebrow labels, account numbers, tickers, tabular figures.* ***Radius:*** *measured* `--radius-sm|md|lg|xl` *= 6/8/12/20px (cards 12px, buttons 8px — both products' actual app-level values, not either's marketing-only pill).*

## 4. Components & Screen Styles

*The component inventory and the styles that realize it, built on the tokens. Styles live in screen.css; full states-per-component table lives in design-system-spec.md § Component inventory.*

| Component | shadcn-svelte primitive | polycarpic usage |
|---|---|---|
| Net-worth hero | Chart (LayerChart, Area) | Big figure + signed delta on the figure + single-line area chart — direct Monarch pattern |
| Grouped account list | Table / custom | Type-grouped rows with sparkline + freshness — direct Monarch pattern |
| Segmented bar + legend | Custom, tokens-driven | Assets/Liabilities, allocation actual-vs-target — direct Monarch pattern |
| Badge | Badge | Entity/account/transaction status, link-health — see Financial semantics |
| Transaction table | Table | Account register, exceptions report, lots |
| Accordion | Accordion | Transactions' three urgency-ordered cards (Exceptions/Pending/Posted); Pending row expansion |
| Near-full-screen modal | Dialog | Journal-entry modal (header + legs) — manual entry, Correct entry, Split, Group & Summarize |
| Entity switcher | Tabs / Sidebar | Switch between Person/Trust/Business/Household |
| Transaction | Sheet | Drill into a posted/draft entry, its journal lines, source doc |

| Component | shadcn-svelte primitive | polycarpic usage |
|---|---|---|
| detail | | |
| Empty state | Card + plain text, no skeleton | Dashboard panels not yet shipped (tax at 1.0, allocation at M4) — see styled-screens/dashboard.html |
| Top row | —, starts right of the sidebar | Not sticky, same background as content: breadcrumb left (expand/collapse lives in the sidebar brand row); entity selector + separator + "more" menu right — AI/search/notifications/gear dropped — see design-system-spec.md § Header |
| Sidebar | Sidebar (shadcn-svelte sidebar-07 block) | Floating card, gutter + rounded all 4 corners, seven nav items, brand top item carrying the expand/collapse control ( `panel-left-close` right-aligned; collapsed, the mark swaps to `panel-left-open` on hover/focus), avatar-popover bottom item opening right — see design-system-spec.md § Sidebar |
| "More" menu | Dropdown Menu | Top row, right-justified — per-page global actions (Sync, New entry, Export, ...) |
| Avatar cluster | —, custom | Overlapping initials in the entity selector, max 3 + overflow |
| Avatar popover | Popover / Dropdown Menu | User menu at the sidebar's bottom, opens to the right — profile, Settings (proposed), Notifications (proposed), preferences, theme, keyboard shortcuts, Log out last |
| Landing illustration | —, custom | Home — the mountains illustration under the brand app bar, then headline, subhead and feature cards; not a signed-in screen |
| Switch | Switch | Settings on/off preferences (auto-post matched recurring transactions, reduced motion) |
| Chart-of- | —, custom (nested | General Ledger: group rows with chevrons, account rows link to the register, custodial cross-reference badge |

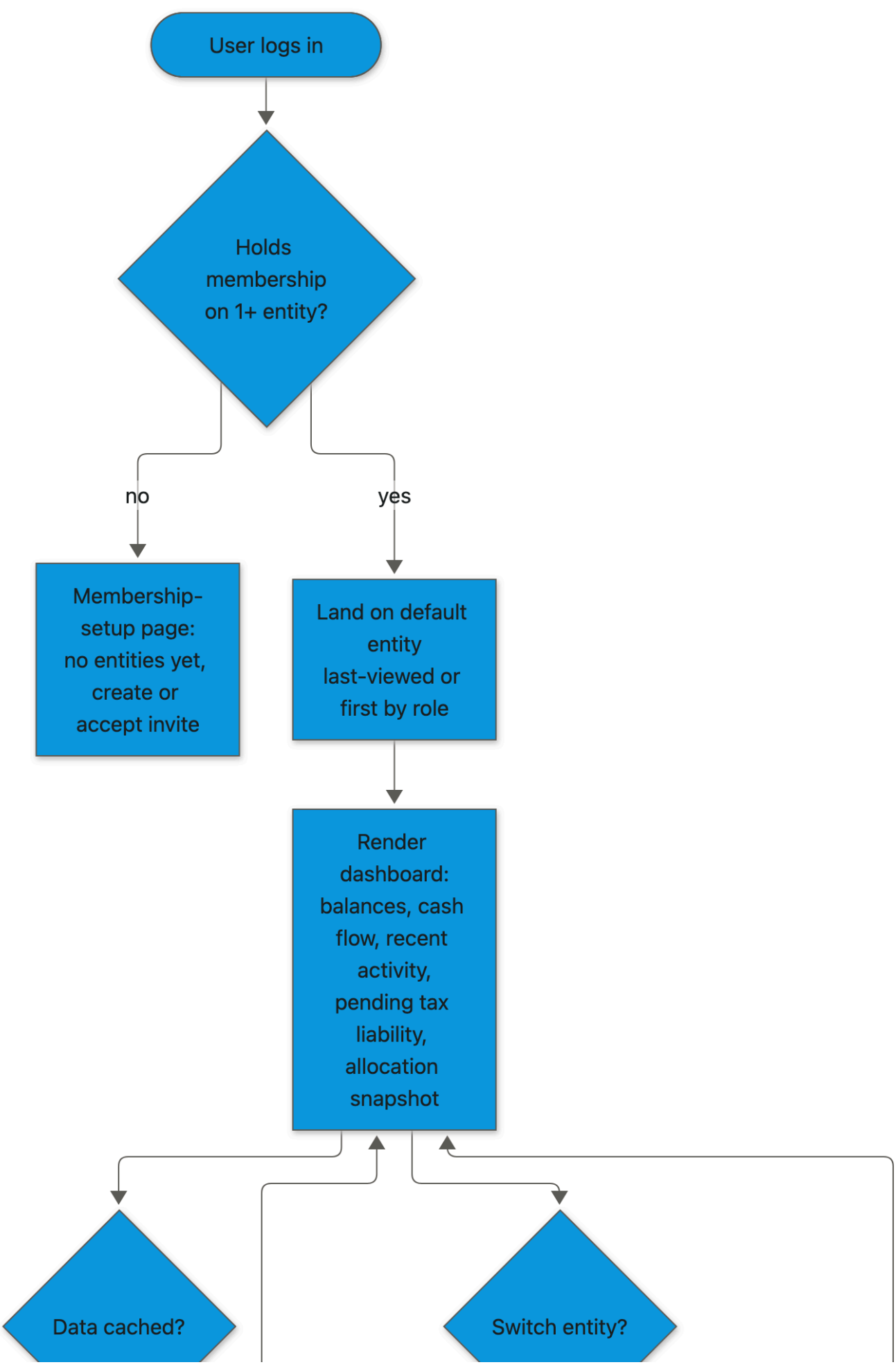
| Component | shadcn-svelte primitive | polycarpic usage |
| --- | --- | --- |
| accounts tree | disclosure rows) | |
| Brand app bar | —, custom | Every page without the app shell (Home, Membership setup, Admin): fixed near-black bar, neon lockup left, the page's trailing items right, a fade below (deep over Home's image, a short shadow over plain pages) |
| Split screen login | —, custom | Login — full field list, centered form column, always-branded visual pane |

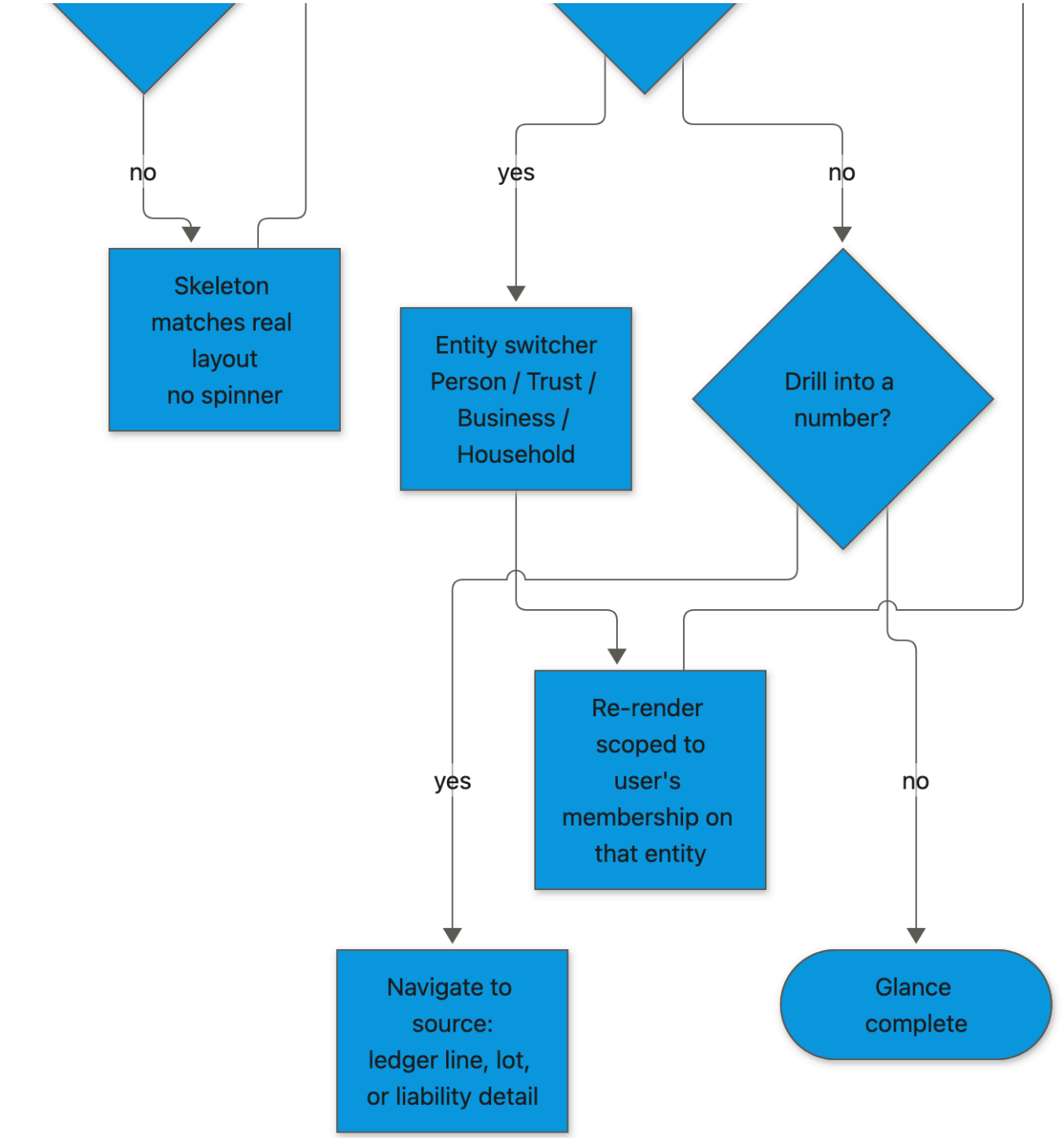
# 5. User Flows

*How users move through the product to accomplish the highest-traffic stories (PRD §4). Each flow is embedded inline below, because a browser serves `flows/*.md` as raw Markdown and its fenced `mermaid` blocks would never reach the page's Mermaid loader. The `.md` files stay as the diffable source of truth in flows/.*

## 5.1 Dashboard glance

***Story (PRD §4.1):*** *per-entity financial health at a glance. Source: flows/dashboard-glance-flow.md.*


User logs in
Holds membership on 1+ entity?
no
yes
Membership-setup page: no entities yet, create or accept invite
Land on default entity last-viewed or first by role
Render dashboard: balances, cash flow, recent activity, pending tax liability, allocation snapshot
Data cached?
Switch entity?

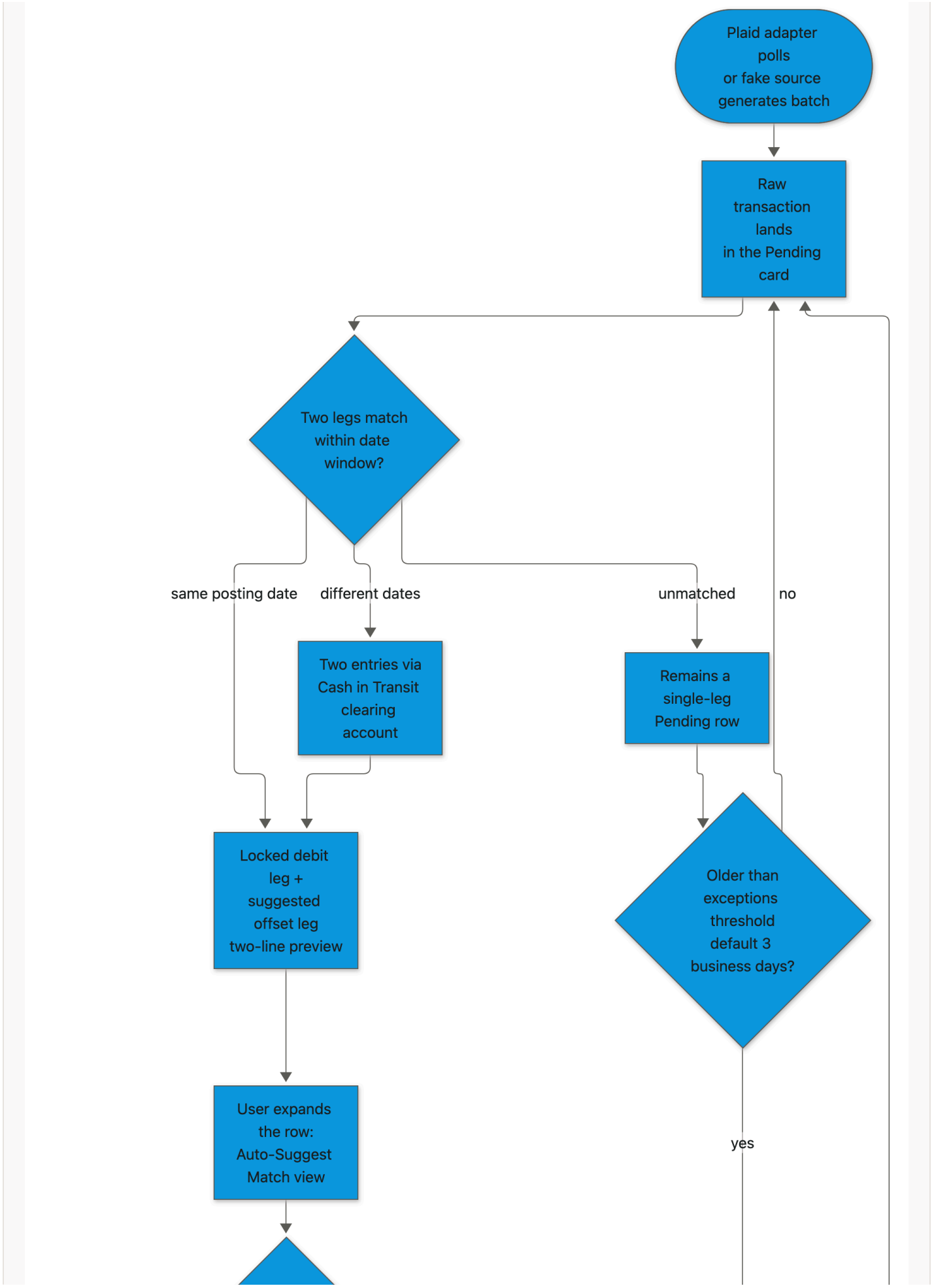


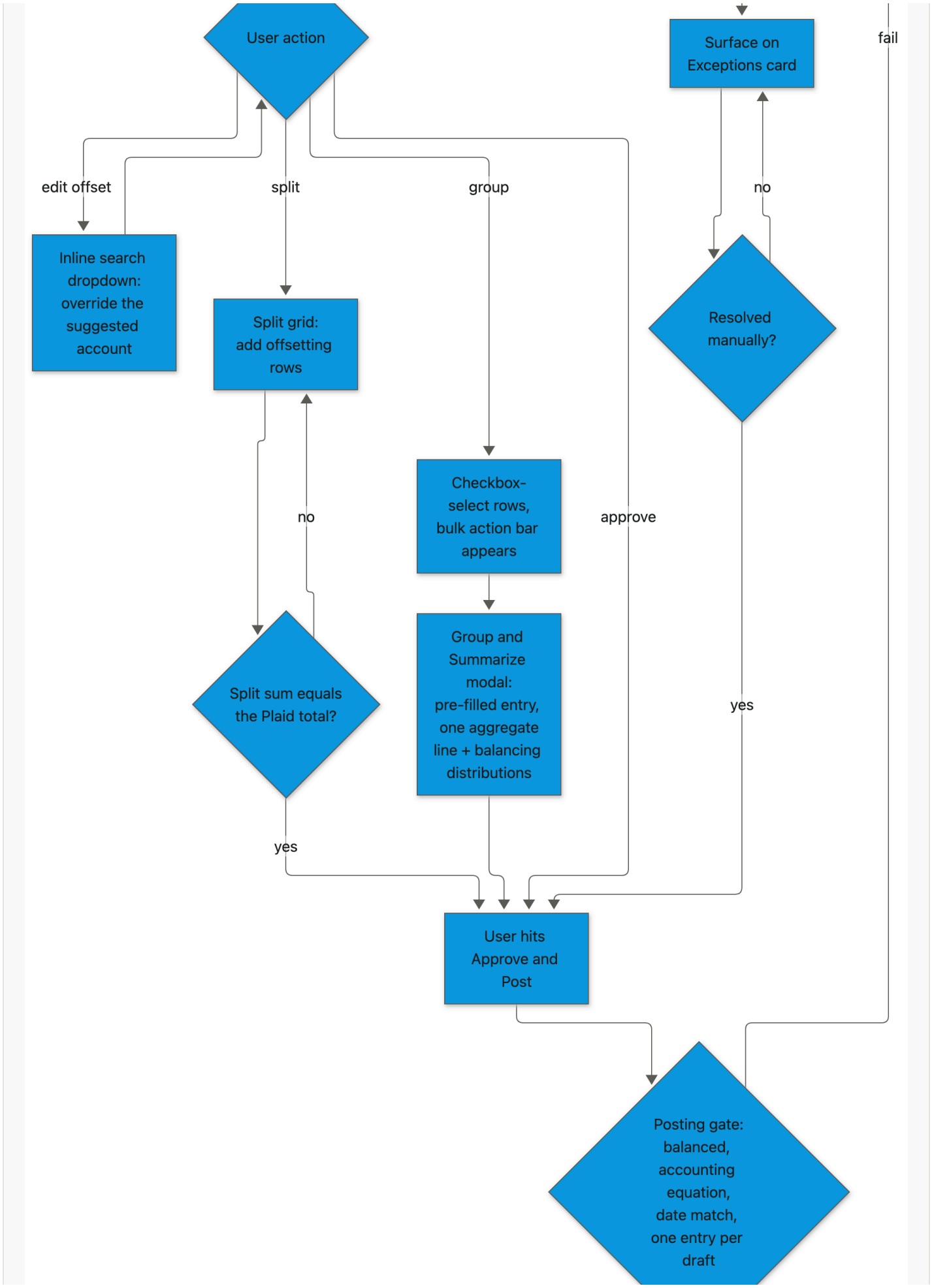

- Every number on this screen must be clickable through to its source (Overview principle: "every number earns trust").
- Filtered strictly by membership (owner/manager/viewer) — no transitive access via parent entity (PRD §5 permission model).
- **No-membership state:** see styled-screens/membership-setup.html. The dashboard's empty state is the membership-setup page itself, not a separate illustrated empty-state screen.
- **"Skeleton matches real layout, no spinner":** while dashboard data is loading, the screen shows placeholder blocks already shaped like the real cards (same card chrome, same

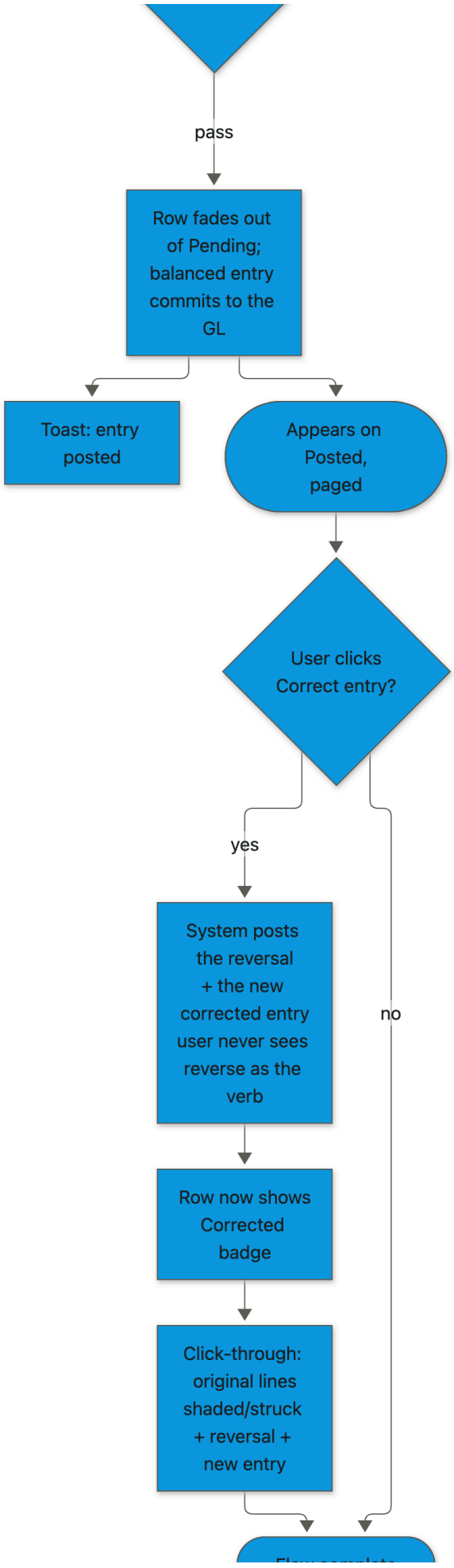
approximate text-line widths) instead of a spinner — so nothing visually jumps or re-flows once the real numbers arrive, they just fade in where the placeholders already were.

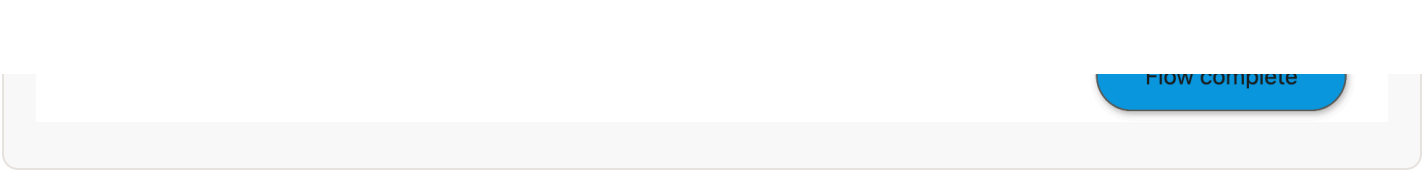
### 5.2 Transactions — review, match & correct

***Story (PRD §4.2):*** *staged for review → matched/split/grouped → posted → correctable. Source: flows/import-confirm-flow.md. The review mechanics below follow a Plaid bank-feed workflow: every imported transaction is reviewed, matched, split or grouped, then posted.*


Plaid adapter polls or fake source generates batch
Raw transaction lands in the Pending card
Two legs match within date window?
same posting date
different dates
unmatched
no
Two entries via Cash in Transit clearing account
Remains a single-leg Pending row
Locked debit leg + suggested offset leg two-line preview
Older than exceptions threshold default 3 business days?
User expands the row: Auto-Suggest Match view
yes


User action
edit offset
split
group
approve
Inline search dropdown: override the suggested account
Split grid: add offsetting rows
Checkbox-select rows, bulk action bar appears
no
Split sum equals the Plaid total?
Group and Summarize modal: pre-filled entry, one aggregate line + balancing distributions
yes
User hits Approve and Post
Surface on Exceptions card
no
Resolved manually?
yes
fail
Posting gate: balanced, accounting equation, date match, one entry per draft

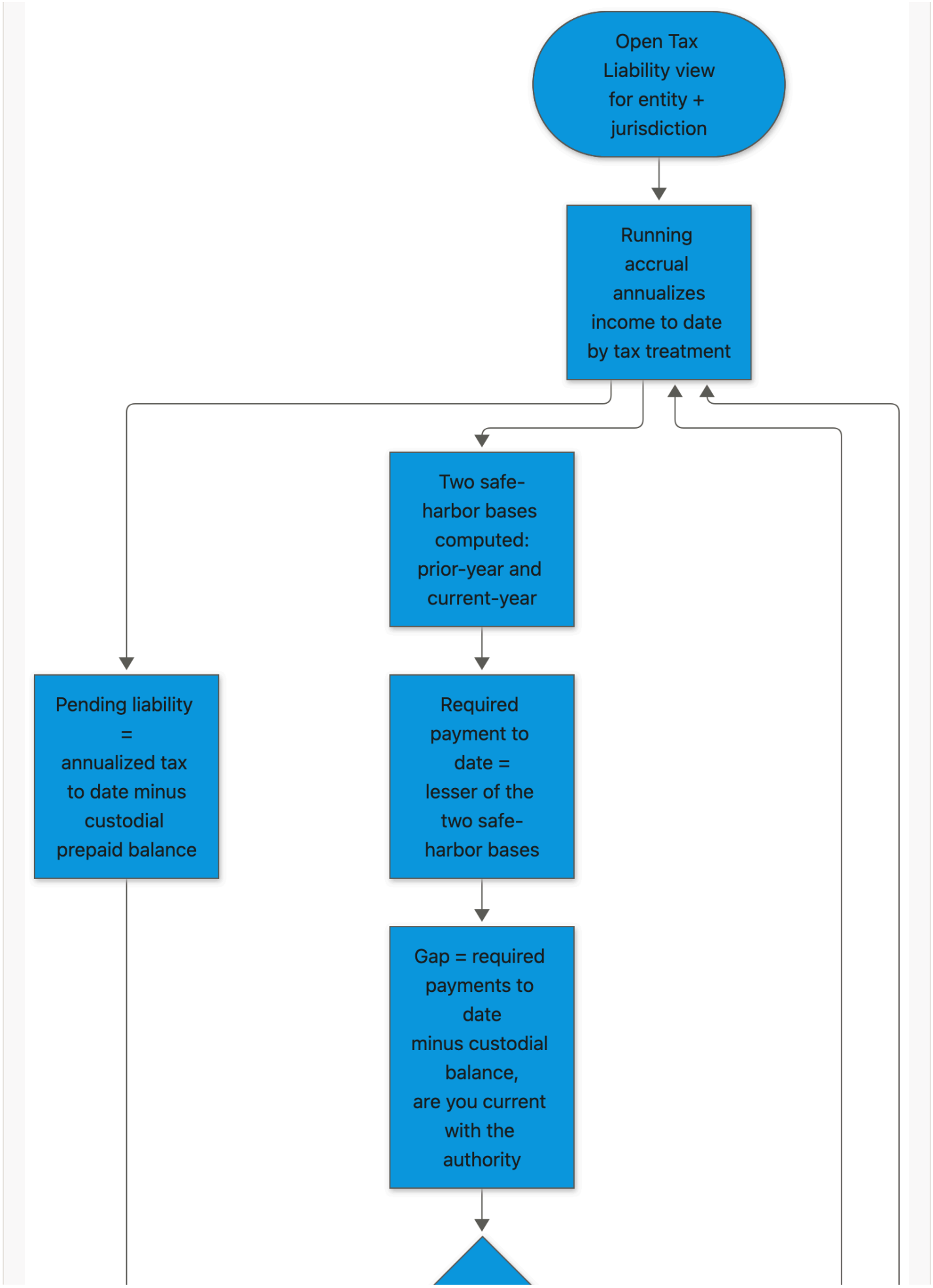

pass
Row fades out of Pending; balanced entry commits to the GL
Toast: entry posted
Appears on Posted, paged
User clicks Correct entry?
yes
no
System posts the reversal + the new corrected entry user never sees reverse as the verb
Row now shows Corrected badge
Click-through: original lines shaded/struck + reversal + new entry

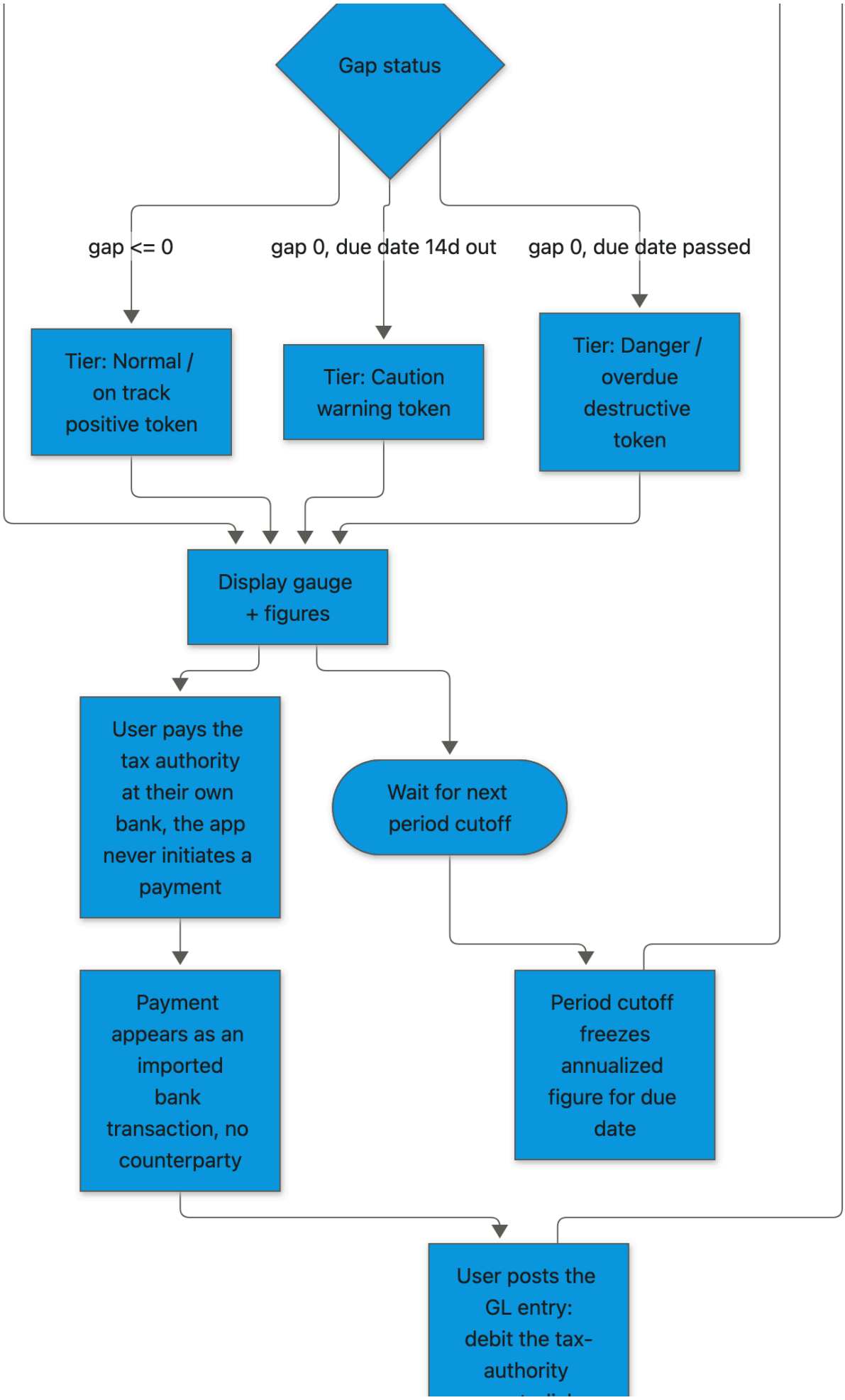


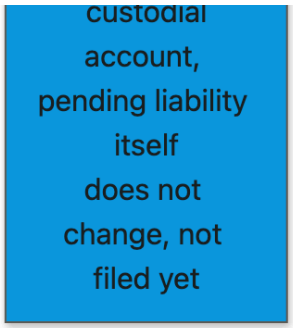

- **The locked leg** is always the custodial cash side (verified bank reality) — never editable, in the preview, Split grid, or Group modal. **The suggested offset leg** is pre-filled from historical rules + Plaid category, overridden in place via an inline search dropdown.
- **Split** is one draft → one multi-leg entry, live-validated to the Plaid total before it's submittable. **Group** is many drafts → one entry (the one many-to-one case, ARCH §3) via the shared journal-entry modal, pre-filled with an aggregate line + balancing distributions.
- **Posting gate:** Σ debits = Σ credits; accounting equation per entity; entry date matches bank date (or the CIT rule); every posted draft linked to exactly one entry — any failure keeps the item in Pending.
- A posted row's user-facing action is "Correct entry", never "edit"/"reverse" — reversal-only underneath (PRD §5). See design-system-spec.md § Correcting a posted entry.
- A corrected row carries the `Corrected` badge ( `--outline` + `--brand-2` -tinted dot, pencil icon); click-through reveals the full double-entry trail — original lines shaded/struck, the reversal, and the new entry. An auto-posted row carries the same badge/action plus a `posted by rule — <name>` line.
- Actor attribution (user / auto-post rule / system job) is shown on every draft and posted row, never inferred (PRD §5).

### 5.3 Tax liability & safe-harbor

***Story (PRD §4.4):*** *pending liability, required payment, gap, pay. Source: flows/tax-liability-flow.md.*


Open Tax Liability view for entity + jurisdiction
Running accrual annualizes income to date by tax treatment
Two safe-harbor bases computed: prior-year and current-year
Pending liability = annualized tax to date minus custodial prepaid balance
Required payment to date = lesser of the two safe-harbor bases
Gap = required payments to date minus custodial balance, are you current with the authority

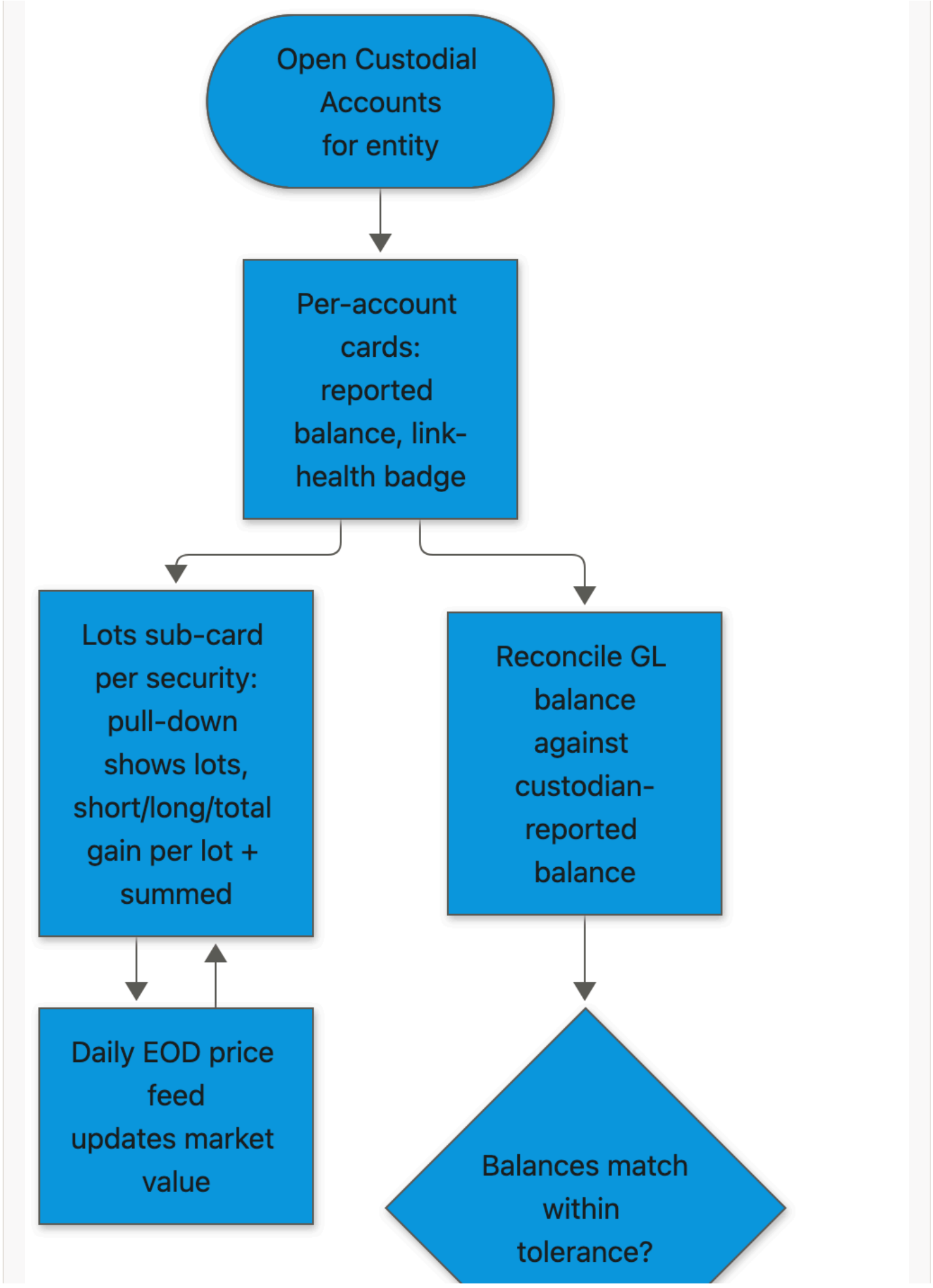

Gap status
gap <= 0
gap 0, due date 14d out
gap 0, due date passed
Tier: Normal / on track positive token
Tier: Caution warning token
Tier: Danger / overdue destructive token
Display gauge + figures
User pays the tax authority at their own bank, the app never initiates a payment
Wait for next period cutoff
Payment appears as an imported bank transaction, no counterparty
Period cutoff freezes annualized figure for due date
User posts the GL entry: debit the tax-authority

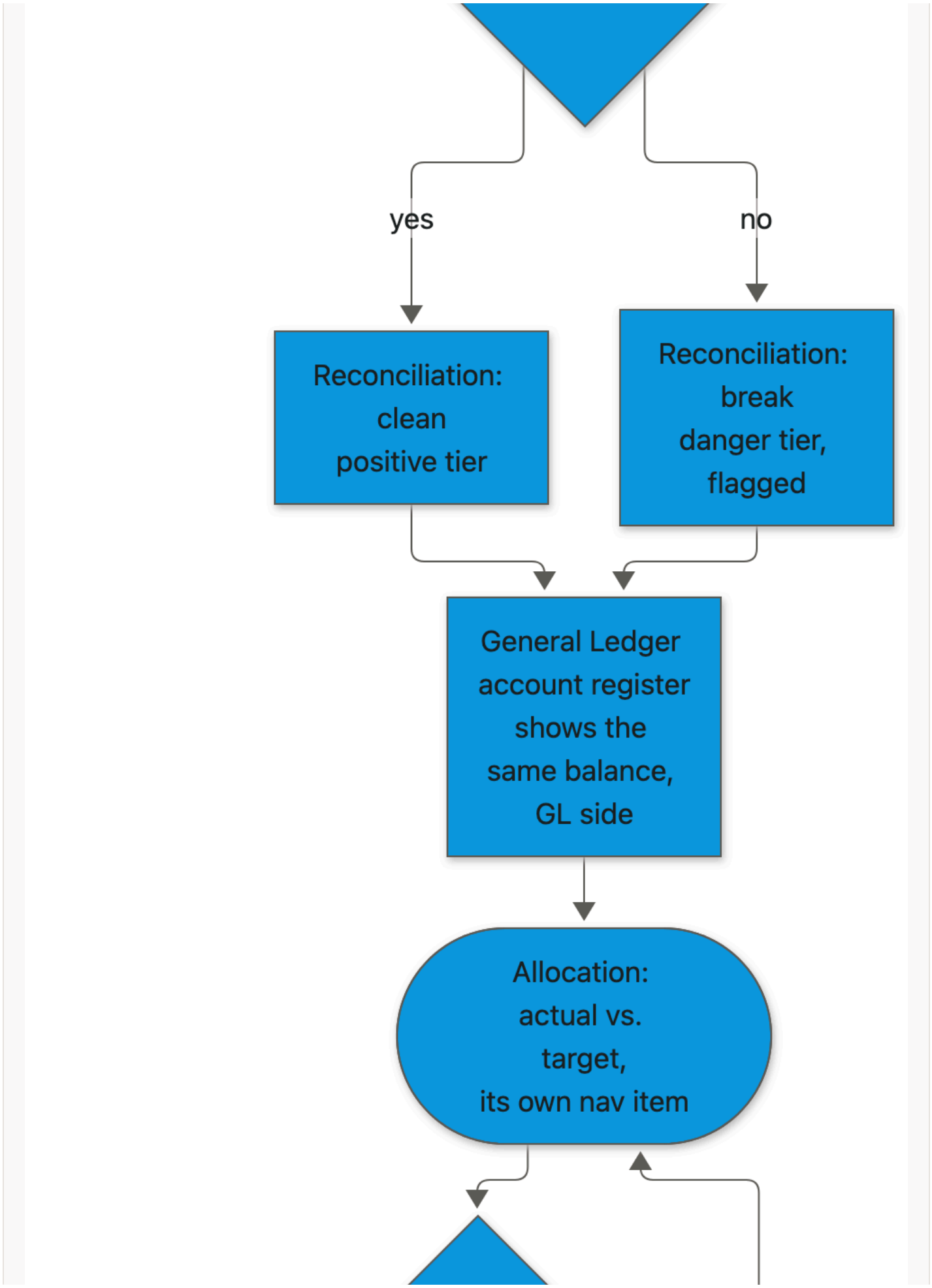


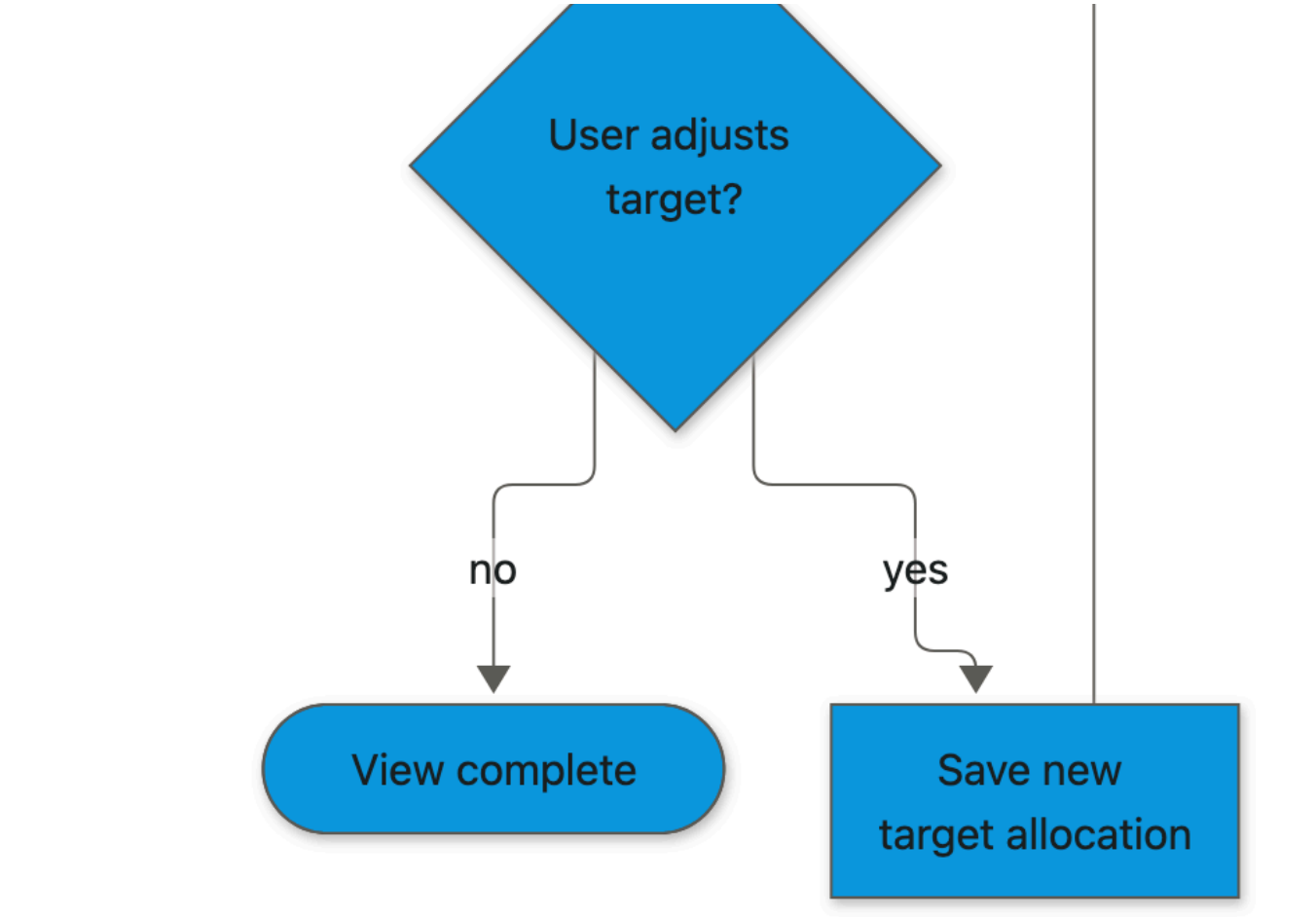

- The three-tier status (normal/caution/danger) reuses the product-wide status vocabulary — see design-system-spec.md § Financial semantics.
- Jurisdiction due dates, multipliers, and cumulative percentages are data-driven, so this view must render correctly for an arbitrary number of jurisdictions (PRD §5).
- **v1 never moves money:** there is no "User initiates payment?" button. The user pays at their own bank; it lands as an imported transaction with no counterparty; the user posts the GL entry, debiting the custodial tax-authority account. Dashboard/tax-panel copy says "post your payment," never "pay" or "transfer."
- **Two figures,** both measured against the custodial account's balance but not against each other — **Pending liability = annualized tax to date – custodial prepaid balance** (what you'd owe net of prepayments, if filed today); **Gap = required payments to date (safe-harbor) – custodial balance** ("are you current with the authority"). Gap is never `required − pending`.

### 5.4 Custodial reconciliation, lots & allocation

***Stories (PRD §4.5–4.7):*** *GL vs. custodial reconciliation, allocation vs. target. Source: flows/securities-lots-flow.md. Lots are a Custodial Accounts sub-card, GL vs. custodial is two nav items rather than a toggle, and Allocation is its own page.*


Open Custodial Accounts for entity
Per-account cards: reported balance, link-health badge
Lots sub-card per security: pull-down shows lots, short/long/total gain per lot + summed
Reconcile GL balance against custodian-reported balance
Daily EOD price feed updates market value
Balances match within tolerance?

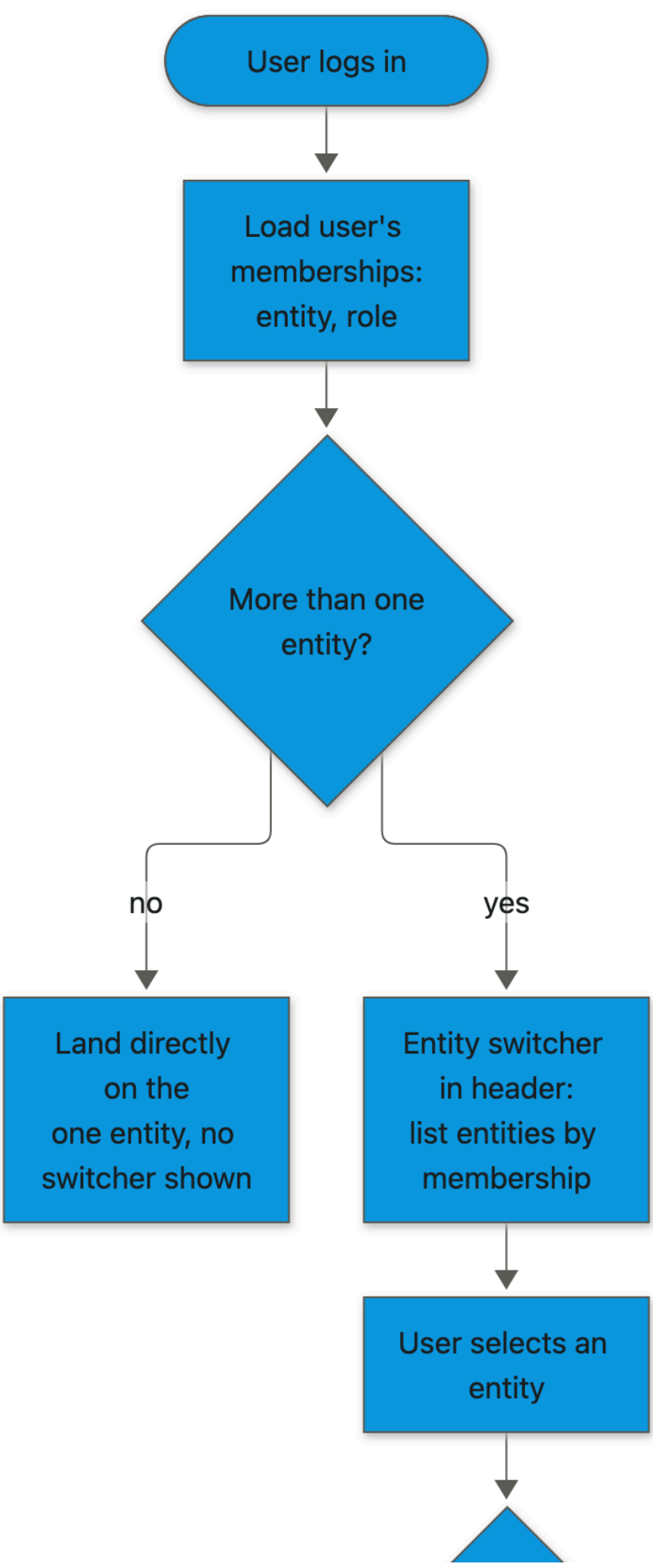

yes
no
Reconciliation:
clean
positive tier
Reconciliation:
break
danger tier,
flagged
General Ledger
account register
shows the
same balance,
GL side
Allocation:
actual vs.
target,
its own nav item

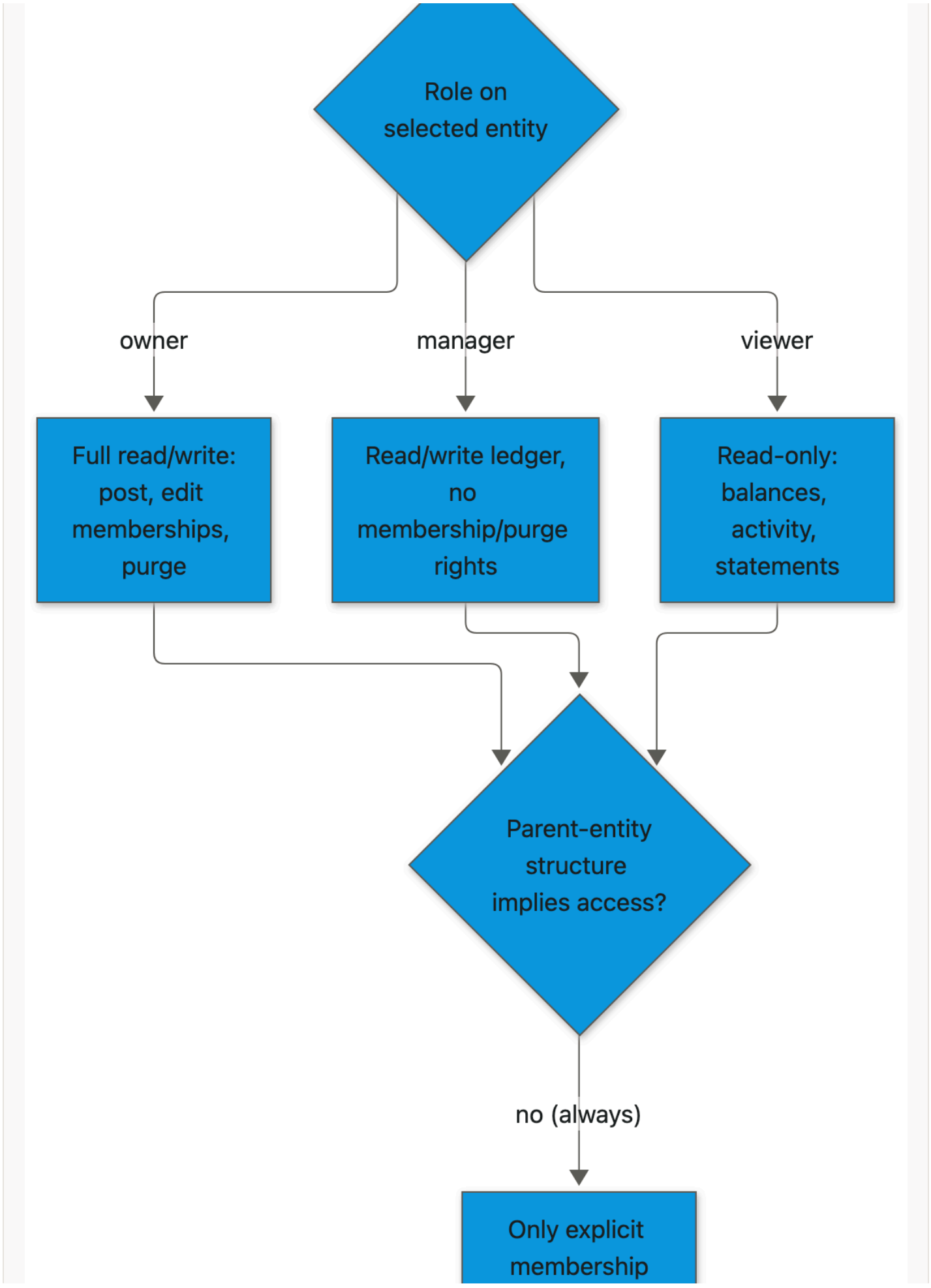


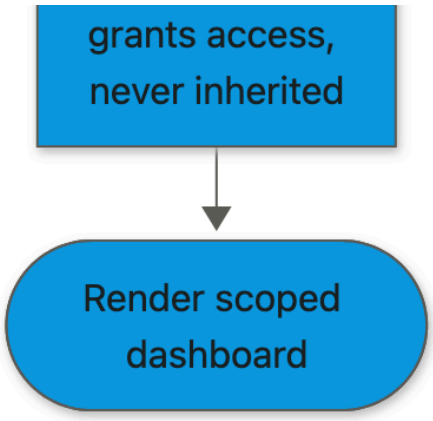

- **No toggle:** General Ledger and Custodial Accounts are two separate nav items, not one screen behind a `Tabs` switch.
- Book-vs-market is implicit on the lots sub-card's own gain/loss figures, not a separate chart.
- A reconciliation break uses the danger tier ( `--destructive` ) and must surface on the Dashboard glance too.

### 5.5 Entity switch & membership access

***Story (PRD §4.8):*** *owner/manager/viewer boundary. Source: flows/entity-membership-flow.md.*


User logs in
Load user's memberships: entity, role
More than one entity?
no
yes
Land directly on the one entity, no switcher shown
Entity switcher in header: list entities by membership
User selects an entity

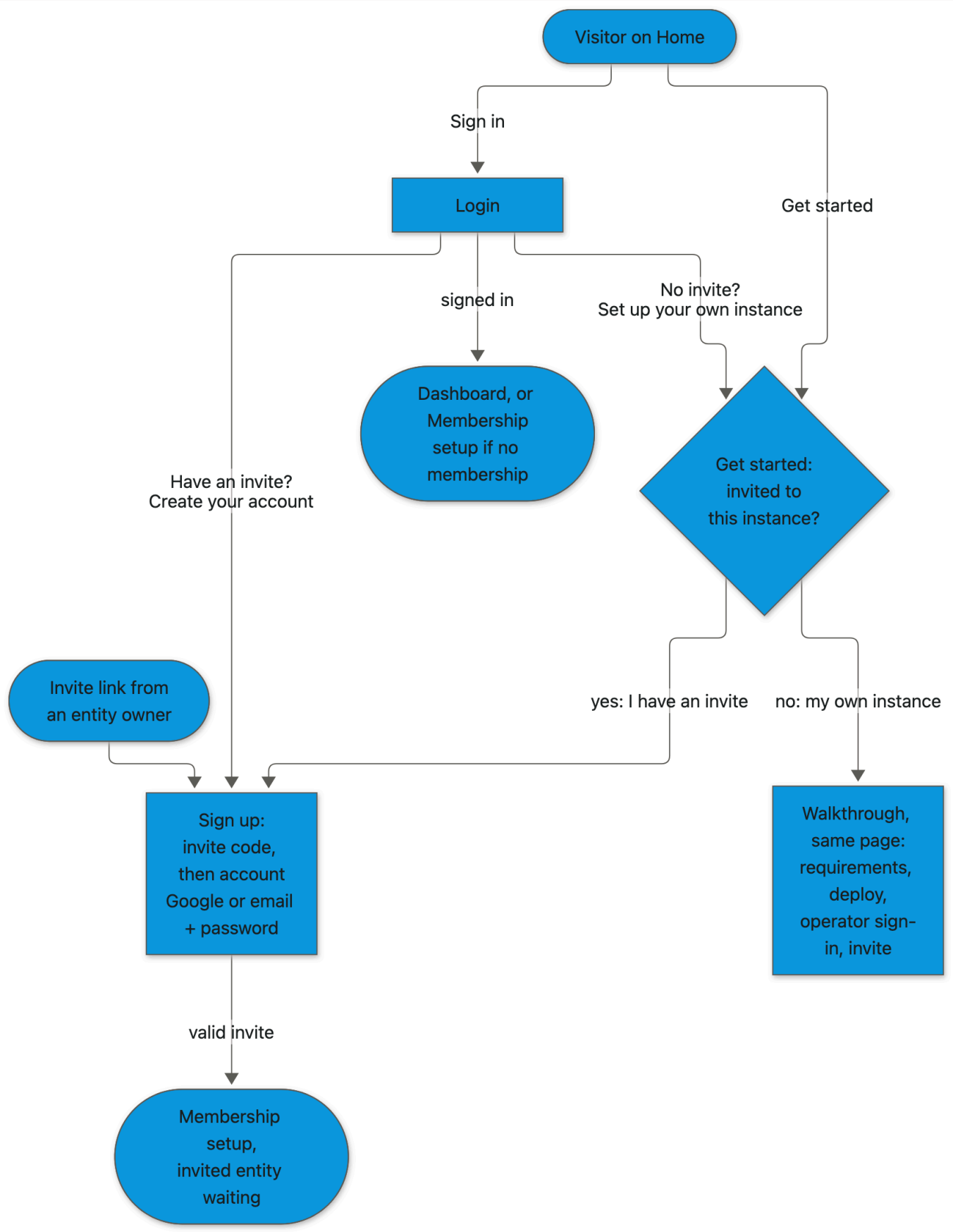

Role on selected entity
owner
manager
viewer
Full read/write: post, edit memberships, purge
Read/write ledger, no membership/purge rights
Read-only: balances, activity, statements
Parent-entity structure implies access?
no (always)
Only explicit membership




- A Household's membership never transitively grants access to a member Person entity, or vice versa (PRD §5, hard requirement).
- A viewer never sees a write affordance rendered-then-disabled — write controls are omitted entirely, not shown-and-greyed.

### 5.6 Home entry: Sign in, Get started, Sign up

***Story:*** *a visitor's first move from Home, on a product with no public sign-up. Source: flows/home-entry-flow.md.*

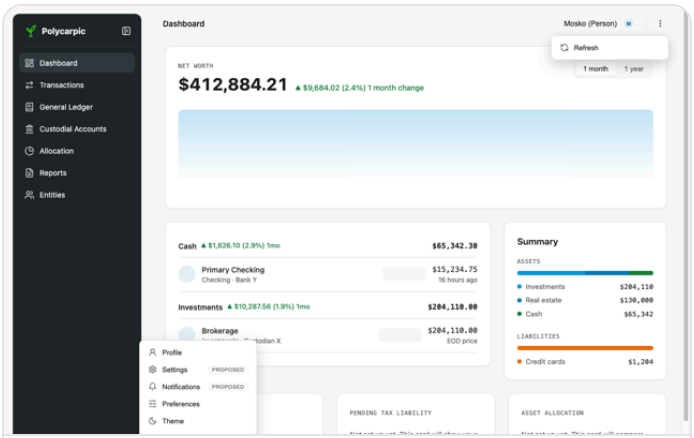

Visitor on Home
Sign in
Login
Get started
signed in
No invite?
Set up your own instance
Dashboard, or
Membership
setup if no
membership
Have an invite?
Create your account
Get started:
invited to
this instance?
Invite link from
an entity owner
yes: I have an invite
no: my own instance
Sign up:
invite code,
then account
Google or email
+ password
Walkthrough,
same page:
requirements,
deploy,
operator sign-
in, invite
valid invite
Membership
setup,
invited entity
waiting

- **No public sign-up:** every instance is self-hosted; accounts come by invitation from an entity owner. The bar's action is "Get started", never "Sign up".
- **Get started is one page:** the two-way fork on top ("I have an invite" → Sign up; "I'm setting up my own instance" → the walkthrough directly below). *Alternative:* a modal from Home's Get started button holding just the two cards; lighter on Home, but the walkthrough then needs its own page anyway.
- **Login shows the same branch:** "Have an invite? Create your account" → Sign up; "No invite? Set up your own instance" → Get started.
- **Sign up is invite-gated:** the code arrives pre-filled from an invite link or is typed; a valid invite names the inviter and the entity; the account (Google, or email + password) hands off to Membership setup. An invalid code shows an inline error, never a dead end.

# 6. Screens

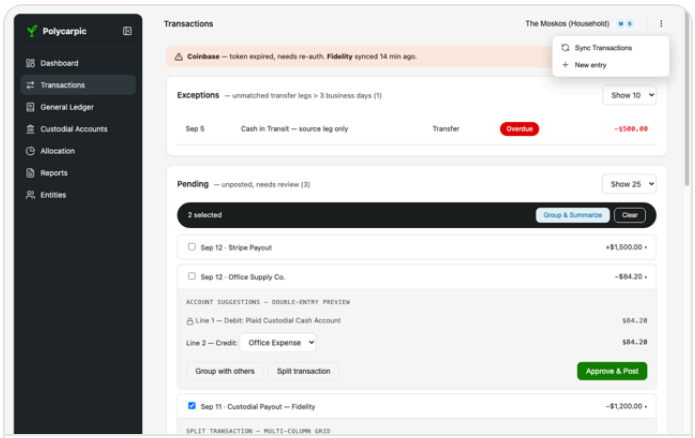
*Every page, rendered with the real design system: `tokens.css` + `screen.css`, the sidebar-07 shell, and Lucide icons. These are the visual-regression reference and the head start for `frontend-lead`. There is one fidelity tier: every page is a styled screen. Each screen supports light and dark (`.dark` on `<html>`) and is checked at 960px and 1280px. Click a screen to open it; the sidebar nav links between them.*

## App screens



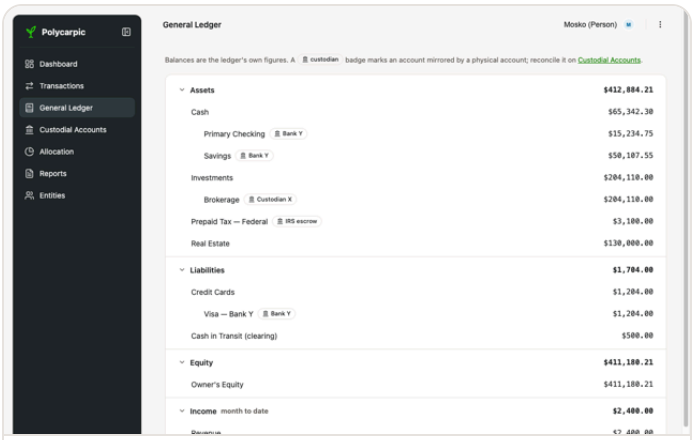

**Dashboard**

Per-entity financial health: net worth, accounts, summary, empty tax/allocation cards.



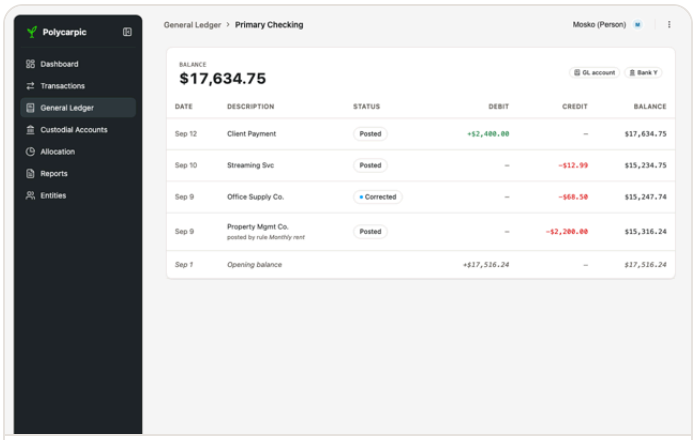

**Transactions**

Exceptions / Pending / Posted accordions; the Plaid review-and-match inbox, split grid, bulk bar, journal-entry modal.



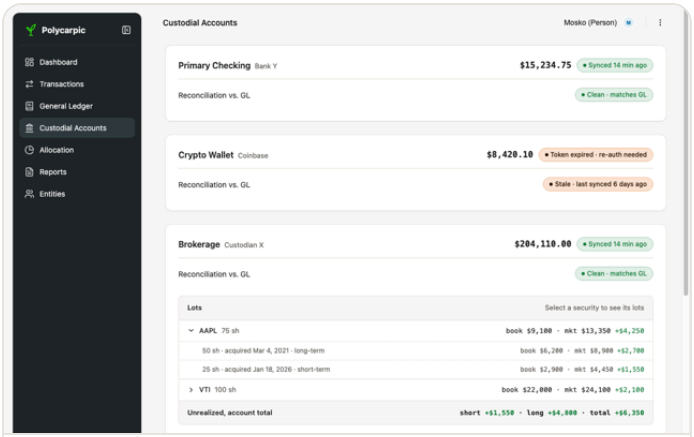

**General Ledger**

Chart-of-accounts tree with custodial cross-reference badges; each account opens its register.



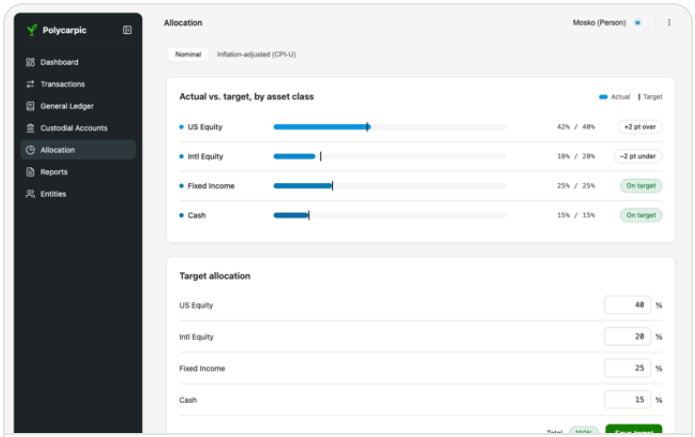

**Account register**

Drill-down from the GL: multi-crumb breadcrumb, Dr/Cr/running-balance table, Corrected and rule-posted rows.



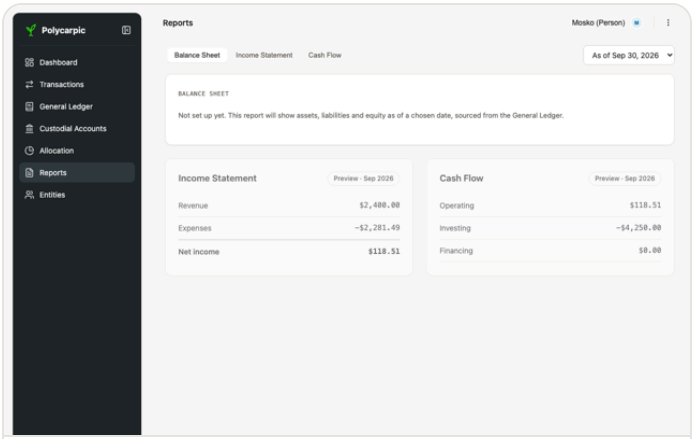

## Custodial Accounts

Per-account cards with link health and reconciliation, a lots sub-card, and the provider-item dialog shown open.



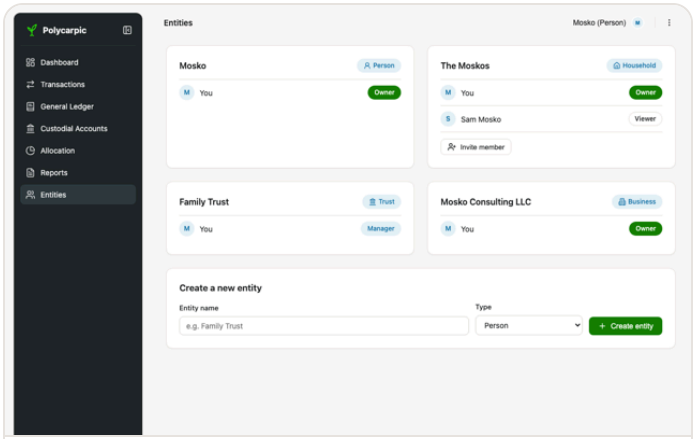

## Allocation

Actual vs. target by asset class with target markers; target editor; nominal / CPI-U toggle.



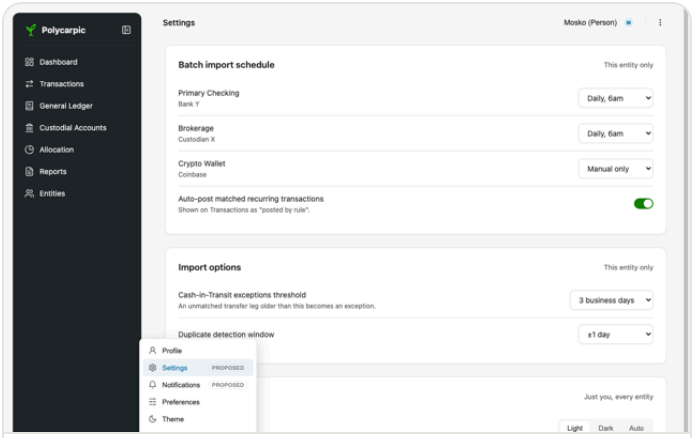

## Reports

Balance Sheet empty state plus dimmed previews of the Income Statement and Cash Flow shapes.




## Entities

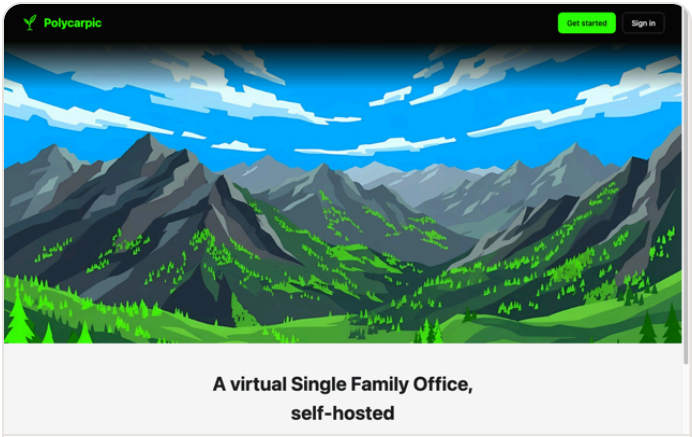
Membership cards with entity-type and role badges, invite, and create.




## Settings

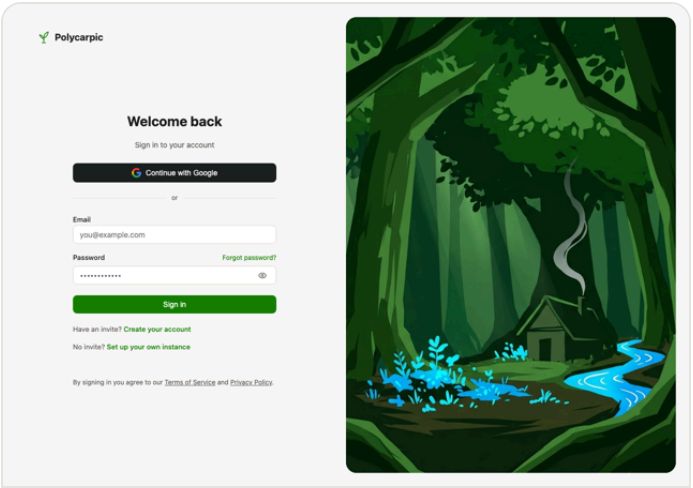
Per-entity import schedule and options, per-user preferences; avatar popover open with Settings current (proposed entry point).

## Public & onboarding



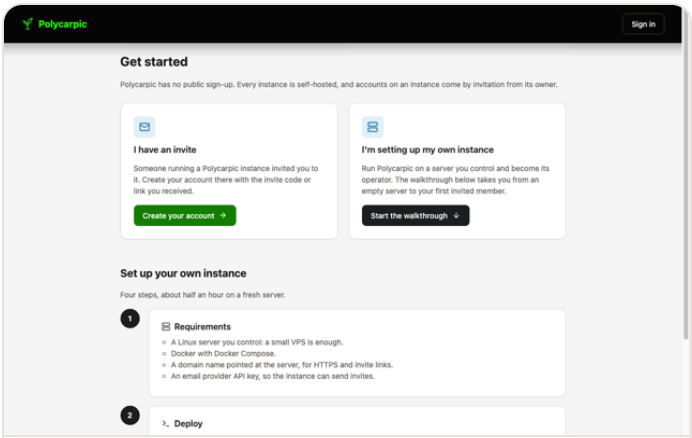

### Home

Public pre-auth landing page: a fixed black top bar (neon lockup left, Get started + Sign in right) fading over the mountains illustration; headline, subhead and cards below.



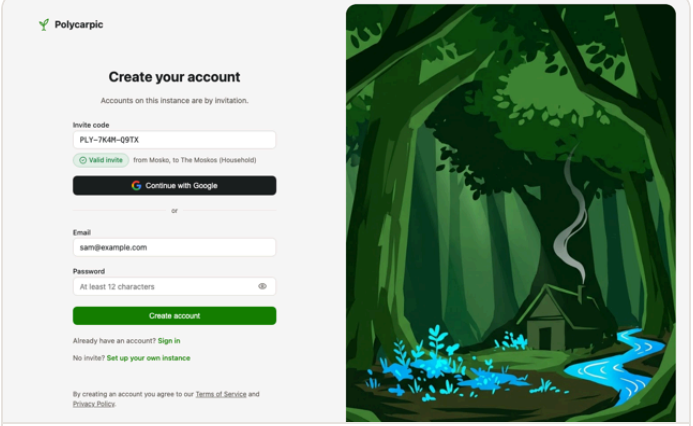

### Login

Split screen: centered form pane, always-branded visual pane.



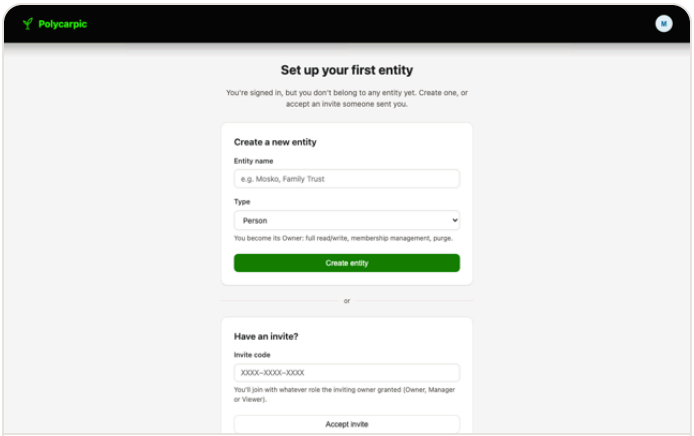

### Get started

Where Home's Get started leads: a two-way fork (I have an invite / I'm setting up my own instance) with the self-host walkthrough below.




### Sign up

Invite-gated account creation in Login's split layout; hands off to Membership setup.

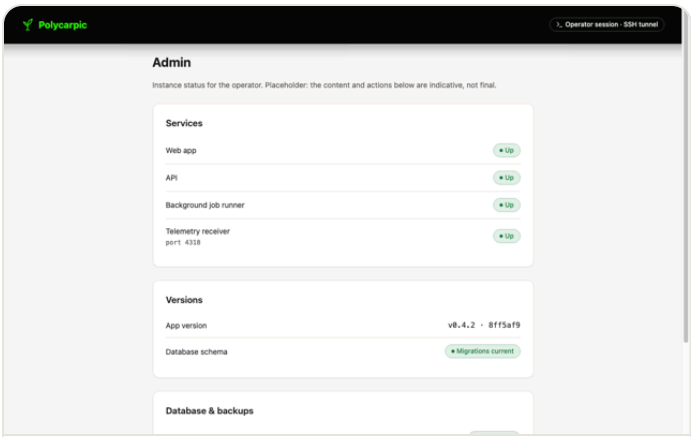



### Membership setup

First run with no membership: create an entity or accept an invite. No sidebar.

## 6.1 Operator surfaces

*Not reachable from any in-app nav and not part of the seven-item IA. Operator-only, reached by SSH tunnel to the host.*




### Admin

**Placeholder, pending PRD/ARCH/SECURITY definition (POLY-81).** Services, versions, database and backups, last sync; administrative actions shown disabled until defined.

## 7. Accessibility

*Accessibility is a design responsibility, not a QA afterthought. Full table in design-system-spec.md § Accessibility.*

| Concern | Standard / approach |
|---|---|
| Color contrast | WCAG AA (4.5:1 text, 3:1 large/UI) — every token pair measured, both modes; `--brand` deliberately fails against light surfaces (1.36:1, by design — never used there, see Foundations); no CVD simulation run yet (open item, high priority given how central the brand hues are) |
| Color-alone signaling | Never — every gain/loss/status pairs color with a sign/icon/label |
| Keyboard | All interactive elements reachable & operable (shadcn default) |
| Screen readers | Semantic markup; labelled controls; delta sign readable from text content |
| Motion | Respect `prefers-reduced-motion` (policy not yet written) |

## 8. Open Questions

*Unresolved design decisions and assumptions that could be wrong. Batched to team-lead for relay to the Principal — not blocking.*

- **Avatar cluster overflow:** the "+N" fold for 4th+ members is specified but not yet shown live in either reference screen (both currently have ≤2 members) — worth a quick visual check once a real multi-member entity exists.
- No CVD (color-vision-deficiency) simulation run on the palette yet — high priority given how central the brand hues are (recommended before lock, design-system-spec.md § Accessibility).
- Native iOS/macOS token export format (kickoff §2.10) — not yet decided, non-blocking for web.
- `prototype/` throwaway UX prototype (kickoff §2.9) not yet built — confirm ownership/timing with team-lead.

## 9. Appendix

- PRD user stories & functional requirements: docs/PRD/index.html §4–5
- Kickoff record, frontend/design system decisions: docs/project_kickoff.md §2.9

### 10.1 FAQs

Q: How do I add review comments to this design doc?

A: Hand-edit `docs/DESIGN/comments.md` with `## §<section-id>` blocks, or run `/serve-docs DESIGN` and use the inline `+ Comment` widget. Then run `/refine-doc DESIGN`. See process/WORKFLOW.md → Doc review loop.

Q: Why is the design system in `docs/DESIGN/` and not in the app's source?

A: It's the design *deliverable* — the contract. During Implement, `frontend-lead` imports or derives the app's real stylesheets from `tokens.css` / `screen.css`.

Q: Where did `variants.css` go?

A: It was generated infrastructure for the `cairn` tracker tool's own theme switcher, never part of the product — removed from this directory (POLY-78 relocates its generator target the same way). See design-system-spec.md § Provenance & shared tooling.

Generated by the `ux-designer` agent via `/generate-designdoc`. Edit by re-running the skill or editing in place.